HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2018)

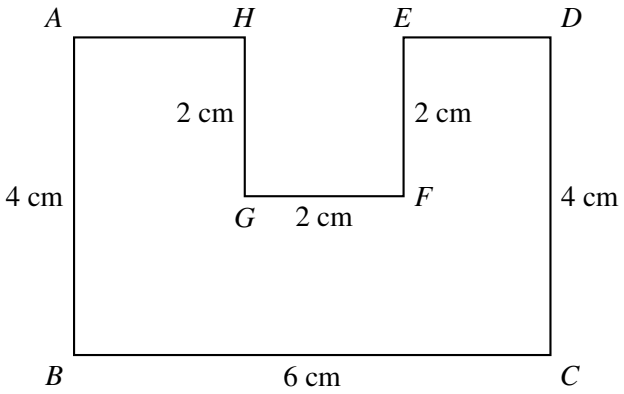
Student No.: 04 Name: _____ Marks: / 22

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, $ABCDEFGH$ is an octagon, where all the measurements are correct to the nearest cm. Let x cm^2 be the actual area of the octagon. Find the range of values of x .



A. $13 < x < 23$

B. $13 < x < 27$

C. $17 < x < 23$

D. $17 < x < 27$

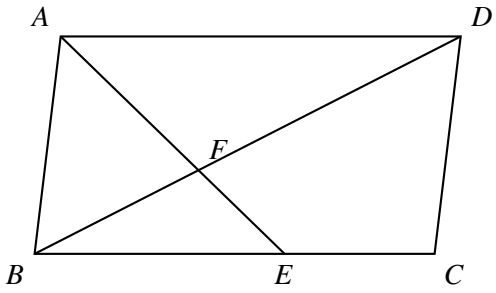
(DSE-2018 PII Q14)

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2. In the figure, $ABCD$ is a parallelogram. E is a point lying on BC such that $BE : EC = 5 : 3$. AE and BD intersect at the point F . If the area of $\triangle ABF$ is 120 cm^2 , then the area of the quadrilateral $CDFE$ is



A. 237 cm^2 .

B. 307 cm^2 .

C. 312 cm^2 .

D. 429 cm^2 .

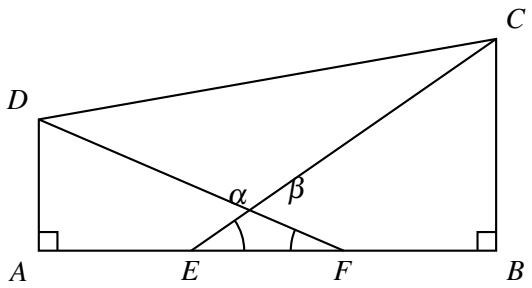
(DSE-2018 PII Q16)

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3. In the figure, $ABCD$ is a trapezium with $\angle ABC = \angle BAD = 90^\circ$. E and F are points lying on AB such that E and F divide AB into three equal parts. Which of the following must be true?



I. $AF \sin\alpha = BE \sin\beta$

II. $CE \cos\alpha = DF \cos\beta$

III. $AD \tan\alpha = BC \tan\beta$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2018 PII Q21)

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4. The polar coordinates of the points C , D and E are $(16, 127°)$, $(12, 217°)$ and $(5, 307°)$ respectively. Find the perimeter of $\triangle CDE$.

A. 54

B. 78

C. 126

D. 130

(DSE-2018 PII Q24)

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5. Two numbers are randomly drawn at the same time from seven cards numbered 1 , 1 , 1 , 2 , 2 , 3 and 4 respectively. Find the probability that the sum of the numbers drawn is 5 .

A. $\frac{5}{21}$

B. $\frac{5}{42}$

C. $\frac{5}{49}$

D. $\frac{10}{49}$

(DSE-2018 PII Q28)

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6. The stem-and-leaf diagram below shows the distribution of the numbers of books read by 20 students in a year.

Stem (tens)	Leaf (units)
2	1 2 2 8
3	a a
4	0 2 4 5 5 7 8
5	3
6	b b 9 9
7	0 8

If the inter-quartile range of the above distribution is at most 25 , which of the following must be true?

I. $5 \leq a \leq 9$

II. $0 \leq b \leq 4$

III. $1 \leq a - b \leq 6$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

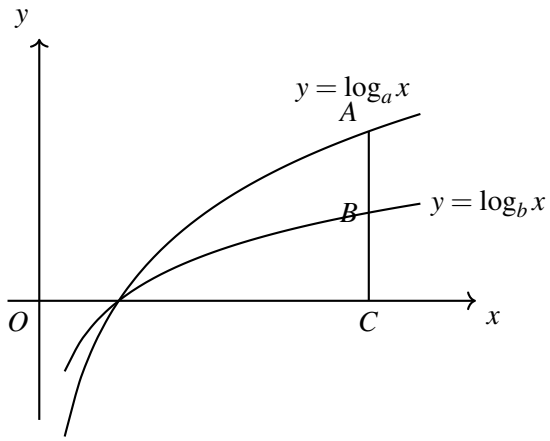
(DSE-2018 PII Q30)

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7. The figure shows the graph of $y = \log_a x$ and the graph of $y = \log_b x$ on the same rectangular coordinate system, where a and b are positive constants. If a vertical line cuts the graph of $y = \log_a x$, the graph of $y = \log_b x$ and the x-axis at the points A , B and C respectively, which of the following is/are true?



I. $a > 1$

II. $a > b$

III. $\dfrac{AB}{BC} = \log_a \dfrac{b}{a}$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2018 PII Q32)

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8. The real part of $\dfrac{2i^{12}+3i^{13}+4i^{14}+5i^{15}+6i^{16}}{1-i}$ is

A. -3 .

B. -1 .

C. 1 .

D. 3 .

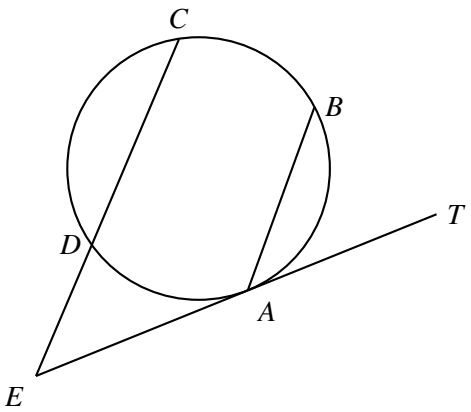
(DSE-2018 PII Q37)

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9. In the figure, TA is the tangent to the circle $ABCD$ at the point A . CD produced and TA produced meet at the point E . It is given that $AB = CD$, $\angle BAT = 24^\circ$ and $\angle AED = 72^\circ$. Find $\angle ABC$.



A. 60°

B. 66°

C. 72°

D. 78°

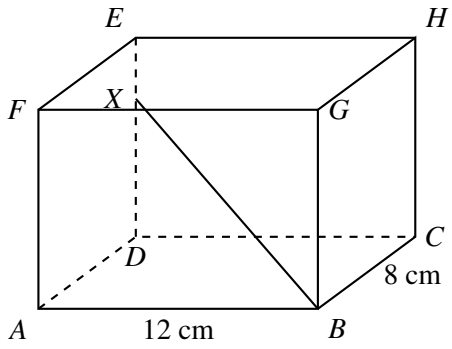
(DSE-2018 PII Q39)

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10. In the figure, $ABCDEFGH$ is a rectangular block. Let X be a point lying on DE such that $DX = 9$ cm and $EX = 4$ cm . Denote the angle between BX and the plane $ABGF$ by θ . Find $\cos\theta$.



A. $\frac{3}{5}$

B. $\frac{4}{5}$

C. $\frac{8}{17}$

D. $\frac{15}{17}$

(DSE-2018 PII Q41)

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11. John and Mary take turns to throw a fair die until one of them gets a number '1' or '6' . John throws the die first. Find the probability that John gets a number '6' .

A. $\frac{1}{2}$

B. $\frac{1}{6}$

C. $\frac{3}{10}$

D. $\frac{7}{10}$

(DSE-2018 PII Q43)

I can finish _____ questions

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